DA6701 – Assignment 4
Portfolio Replication

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1 Introduction

The objective is to replicate the return profile of the S&P 500 index given for a 5 year period, using only a subset of the stocks. Selecting k stocks out of 500 to minimise the variance is an NP-hard problem. To arrive at an approximate solution, we have used: Lasso Regression, and Latent Factor Mapping with Hierarchical Risk Parity.

2 Data Loading

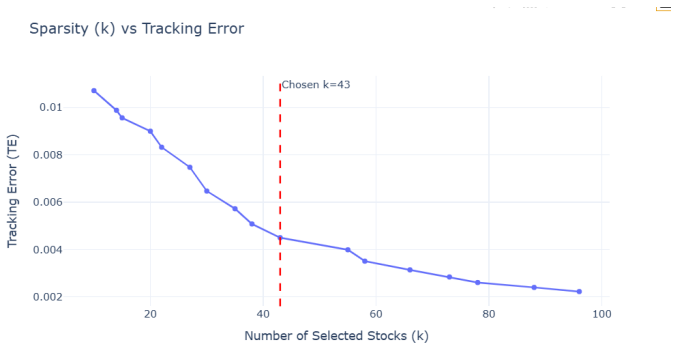
We use constituent stock returns as features and S&P 500 returns as target. We have used Jan 2020 to June 2025 (1326 days) for training the model and July 2025 to Dec 2025 (128 days) to test the model. We have removed the columns which have more than 5% missing entries, apart from other missing entries. In the end, 24 stocks were dropped due to missing entries.

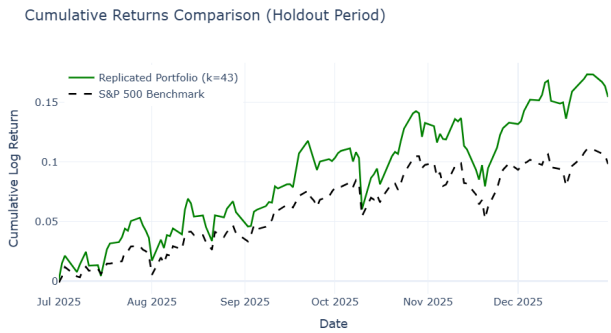
3 Portfolio Replication

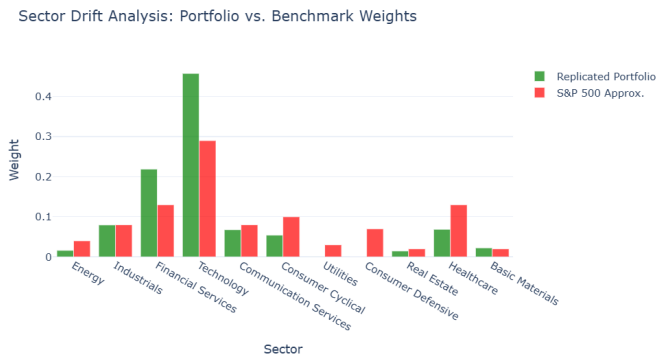
3.1 Lasso Regression

- Fit Lasso Regression on training data without an intercept
- Select top-k stocks based on non-zero weights and normalise them
- Generate a grid of α values
- For each α, build a portfolio without any constraint on k
- Evaluate **Tracking Error** and **Information Ratio** on test set
- Choose α that selects ≤ 50 stocks while minimising Tracking Error

Deliverables:







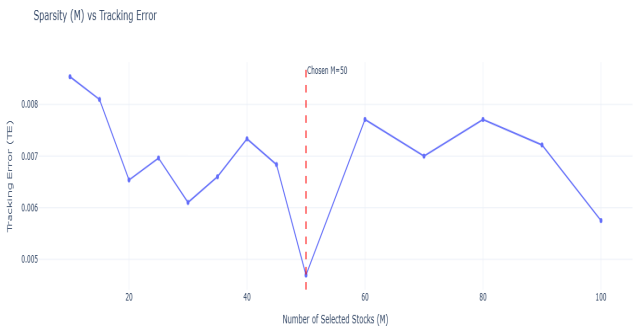
Performance Metrics

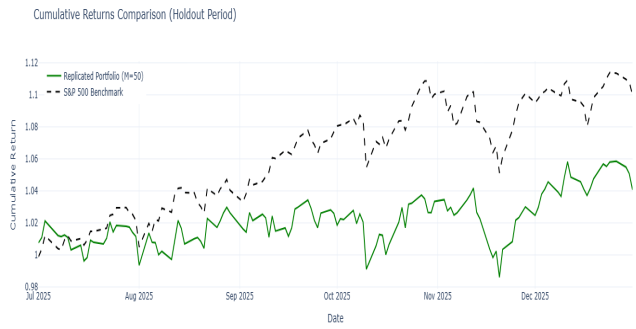
Chosen Alpha		4.5×10^{-5}
Final Count (k)		43
Tracking Error	Train	0.008204
	Test	0.004503
Information Ratio	Train	0.0536
	Test	0.0973

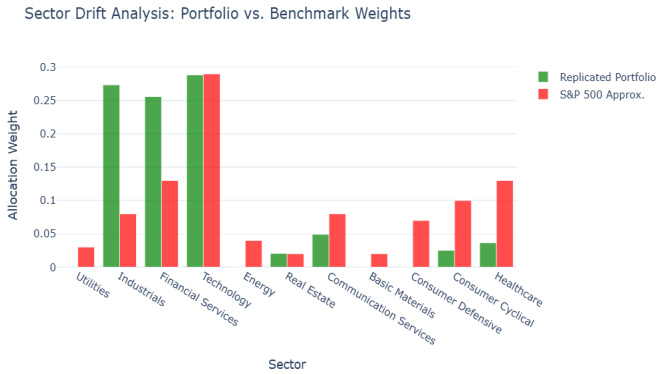
3.2 Latent Factor Mapping

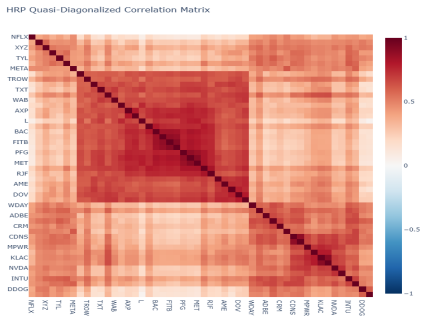
- Use a PyTorch autoencoder to extract k latent factors from stock returns
- Compute correlation between each stock and latent factors
- Select top M/k most correlated stocks per factor
- Assign portfolio weights using HRP (quasi-diagonalisation + recursive bisection)
- Tune hyperparameters k and M to minimise Tracking Error

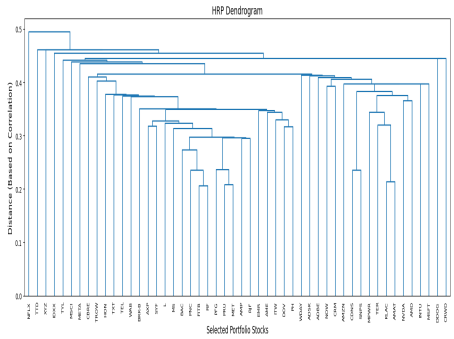
Deliverables:











Performance Metrics

Factors Extracted (k)		2
Final Count (M)		50
Tracking Error	Train	0.004618
	Test	0.004695
Information Ratio	Train	0.0349
	Test	-0.0903

4 Observation

Metric	**Lasso Regression**	**Latent Factor Mapping**
TE Train	0.008204	0.004618
TE Test	0.004503	0.004695
IR Train	0.0536	0.0349
IR Test	0.0973	-0.0903

Table 1: Comparison of Lasso Regression and Latent Factor Mapping

Lasso Regression directly models index returns using stock weights, whereas Latent Factor Mapping uses an autoencoder to extract latent factors and constructs a portfolio through correlation-based selection and HRP. Both methods achieve comparable Tracking Error, with Lasso performing slightly better on the test set.

However, Lasso significantly outperforms in terms of Information Ratio, indicating superior risk-adjusted performance and better generalisation. In contrast, Latent Factor Mapping exhibits a negative IR on test data, likely due to instability introduced by its multi-stage pipeline (autoencoder, correlation-based selection, and HRP) and its lack of direct optimisation for Tracking Error. Overall, Lasso provides a more stable, interpretable, and effective portfolio construction method.

5 Conclusion

We compared Lasso Regression and Latent Factor Mapping for portfolio replication under sparsity constraints, with Lasso demonstrating superior performance.
Note: Please refer to the visualizations/ for detailed deliverables